Adjacent vertices fault-tolerance for bipancyclicity of hypercube*

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Abstract

A bipartite graph $G = (V, E)$ is bipancyclic *if it contains the cycles of every even length from 4 to $|V|$. Let F_a be the set of f_a pairs of adjacent vertices and F_e be the set of f_e faulty edges in the n-dimensional hypercube Q_n. In this paper, we will show that $Q_n - F_a - F_e$ is bipancyclic for $f_a + f_e = n - 2$.*

1 Introduction

It is well known that the hypercube network is one of the most popular and efficient topological structures of interconnection networks. This is partially due to its many excellent features, such as regularity, symmetry, low diameter and degree, high connectivity, simple routing algorithms, and so on. The ring network is the most fundamental interconnection networks. It is suitable for designing simply algorithms with low cost. Component failures are unavoidable in a large parallel systems. Thus, fault tolerance is an important research issue for interconnection networks.

The interconnection network is usually represented by a simple undirected graph where vertices represent processors and edges represent links between processors. Let $G = (V_b \cup V_w, E)$ be a *bipartite graph* where V_b and V_w are two disjoint vertex sets such that each edge of E consists of one vertex from each set. For convenient, we call V_b black vertex set and V_w white vertex set. Two vertices u and v are adjacent if $(u, v) \in E$. A *path* is a sequence of adjacent vertices, denoted as $\langle v_1, v_2, \cdots, v_n\rangle$, where all vertices are distinct. We also denote this path as $\langle v_1, v_2, \cdots, v_i \overset{P(v_i,v_j)}{\longrightarrow} v_j, \cdots, v_n\rangle$ where $P(v_i, v_j)$ is the subpath $\langle v_i, v_{i+1}, \cdots, v_j\rangle$. A *Hamiltonian path* is a path that spans G. A *cycle*, written as $\langle v_1, v_2, \cdots, v_n\rangle$, is a path for $v_1 = v_n$. The *length* of a path or a cycle is the number of edges in it. A *Hamiltonian cycle* is a cycle that visits every vertex exactly once. A *Hamiltonian graph* is a graph that contains a Hamiltonian cycle. A graph $G = (V, E)$ is k edges Hamiltonian if $G - F_e$ is a Hamiltonian graph for every $F_e \subset E$ and $|F_e| \leq k$. A bipartite graph $G = (V_b \cup V_w, E)$ is *Hamiltonian laceable* if there exists a *Hamiltonian path* between x, y for any $x \in V_b, y \in V_w$. A bipartite graph $G = (V_b \cup V_w, E)$ is *k edges Hamiltonian laceable* if $G - F_e$ is Hamiltonian laceable for every $F_e \subset E$ and $|F_e| \leq k$. A bipartite graph $G = (V_b \cup V_w, E)$ is *bipancyclic* if it contains the cycles of every even length from 4 to $|V(G)|$ inclusive. A bipartite graph $G = (V_b \cup V_w, E)$ is *k edges bipancyclic* if $G - F_e$ is bipancyclic for every $F_e \subset E$ and $|F_e| \leq k$. A bipartite graph $G = (V_b \cup V_w, E)$ is *edge-bipancyclic* if every edge of G lies on cycles of every even length from 4 to $|V(G)|$. A bipartite graph $G = (V_b \cup V_w, E)$ is *k edges edge-bipancyclic* if $G - F_e$ is edge-bipancyclic for every $F_e \subset E$ and $|F_e| \leq k$.

An n-dimensional hypercube $Q_n = (V_b \cup V_w, E)$ is a bipartite graph whose vertices are labeled by distinct n-bit binary strings. Two vertices are linked by an edge if and only if their labels differ exactly in one bit.

Tsai et al. proved that Q_n is $(n - 2)$ edges Hamiltonian laceable in [5]. In [3], Li et al. proved that Q_n is $(n - 2)$ edges bipancyclic. Hsieh et al.

*This work was supported in part by the National Science Council of Republic of China under NSC 95-2221-E-212-028-

showed that every edge in $Q_n - F_v - F_e$ lies on cycles of every even length from 4 to $2^n - 2|F_v|$ if $|F_v| + |F_e| \leq n - 2$ in [1]. However, there may exists some cycle with length more than $2^n - 2|F_v|$ in $Q_n - F_v - F_e$ for $|F_v| + |F_e| \leq n - 2$. This result is not the vertex fault tolerance for bipancyclic of Q_n.

Hung et al. investigated the adjacent vertices fault tolerance Hamiltonian laceable for hypercube in [2]. A bipartite graph $G = (V_b \cup V_w, E)$ is *f_a-adjacency f_e edges Hamiltonian* if $G - F_a$ is f_e edges Hamiltonian $\forall F_a = \{a_i, b_i | (a_i, b_i) \in E$, for $1 \leq i \leq f_a\}$. A bipartite graph $G = (V_b \cup V_w, E)$ is *f_a-adjacency f_e edges Hamiltonian laceable* if $G - F_a$ is f_e edges Hamiltonian laceable $\forall F_a = \{a_i, b_i | (a_i, b_i) \in E$, for $1 \leq i \leq f_a\}$. They proved that Q_n is f_a-adjacency $(n - 2 - f_a)$ edges Hamiltonian for $1 \leq f_a \leq (n - 2)$ and f_a-adjacency $(n - 2 - f_a)$ edges Hamiltonian laceable for $0 \leq f_a \leq (n - 3)$ in [2].

In this paper, we investigate the adjacent vertices fault tolerance for bipancyclicity of hypercube. A bipartite graph $G = (V_b \cup V_w, E)$ is *f_a-adjacency f_e edges bipancyclic* if $G - F_a$ is k edges bipancyclic $\forall F_a = \{a_i, b_i | (a_i, b_i) \in E$, for $1 \leq i \leq f_a\}$. We will show that Q_n is f_a-adjacency $(n-2-f_a)$ edges bipancyclic for $0 \leq f_a \leq (n-2)$.

The rest of this paper is organized as follows. In Section 2, we will prove adjacent vertices fault tolerance for bipancyclicity of hypercube. Some conclusions are given in Section 3.

2 The adjacent vertices fault tolerance for bipancyclicity of hypercube

In this section, we will prove the adjacent vertices fault tolerance for bipancyclicity of hypercube. Before the proof of this resnlt, we need some properties for hypercube.

The following lemma was proved by Hung et al. in [2].

Lemma 1 *For $n \geq 3$, Q_n is f_a-adjacency $(n - 2 - f_a)$ edges Hamiltonian for $0 \leq f_a \leq (n - 2)$ and f_a-adjacency $(n - 2 - f_a)$ edges Hamiltonian laceable for $0 \leq f_a \leq (n - 3)$.*

Let $G = (V, E)$ be a graph and s_1, s_2, t_1, t_2 be some vertices of G. The two paths $P(s_1, t_1)$ and $P(s_2, t_2)$ are two spanning disjoint paths if every vertex of V is either on $P(s_1, t_1)$ or $P(s_2, t_2)$ and not both. Park et al. proved the following Lemma in [4].

Lemma 2 *Let $Q_n = (V_b \cup V_w, E)$ be the n-dimensional hypercube. For any $s_1, s_2 \in V_b$ and $t_1, t_2 \in V_w$, there exist two spanning disjoint paths $P_1(s_1, t_1)$ and $P_2(s_2, t_2)$ in Q_n.*

In [3], Li et al. proved the following Lemma.

Lemma 3 *The graph Q_n is $(n - 2)$ edges edge-bipancyclic.*

We will prove the adjacent vertices fault tolerance for bipancyclicity of hypercube. We need some definitions in the following proof. An i-edge (x, y) is an edge that x and y differ in the i-th bit. And let $dim((x, y)) = i$. The hypercube Q_n can be constructed recursively as $Q_n = Q_{n-1} \times K_2$. We can partition Q_n as two subgraphs Q^0_{n-1} and Q^1_{n-1} by choosing any one bit of binary string. Let S be an edge subset of Q_n and $DIM(S) = \{dim(e) | e \in S\}$.

Theorem 1 *The hypercube Q_n is f_a-adjacency $(n - 2 - f_a)$ edges bipancyclic for $0 \leq f_a \leq n - 2$ and $n \geq 2$.*

Proof. Let F_a be the set of f_a pairs of adjacently faulty vertices of Q_n and F_e be the set of faulty edges for $|F_e| = f_e = n-2-f_a$. We use $DIM(F_a)$ to denote the set of dimension of the edge between every pair of adjacently faulty vertices. We will construct the cycles with every even length from 4 to $2^n - 2f_a$ in $Q_n - F_e - F_a$ with $f_a + f_e = n - 2$. We will prove this theorem with induction on n. Let x be a vertex of Q_n and x_i be the ith bit of x. We can choose any bit i such that the induced subgraph of the vertices x with $x_i = k$ in Q_n is isomorphic to Q_{n-1} for $k = 0, 1$ and $1 \leq i \leq n$. Since $f_a + f_e = n - 2$, there exists a bit j for $j \notin (DIM(F_a) \cup DIM(F_e))$. We use Q^0_{n-1} to denote the subgraph of $Q_n = (V, E)$ induced by $\{x \in V | x_j = 0\}$ and Q^1_{n-1} to denote the subgraph of Q_n induced by $\{x \in V | x_j = 1\}$. We use $\phi(x) \in$

$V(Q_{n-1}^1)$ to denote the vertex adjacent to x for any vertex $x \in V(Q_{n-1}^0)$. Let f_a^k be the number of pairs of adjacently faulty vertices and f_e^k be the number of faulty edges in Q_{n-1}^k for $k = 0, 1$. Thus, $f_a = f_a^0 + f_a^1$ and $f_e = f_e^0 + f_e^1$.

Applying Lemma 3, we can obtain that this theorem is true for $f_a = 0$. This theorem is trivial true for $n \leq 3$. In the following, we will assume that $f_a \geq 1$ and $n \geq 4$. Without loss of generality, we can assume that $f_a^0 \geq f_a^1$. We will prove this theorem with the following cases.

Case 1: $f_a^1 = f_e^1 = 0$

Applying Lemma 3, Q_{n-1}^1 is bipancyclic. Thus, there exist cycles with every even length from 4 to 2^{n-1}. With induction hypothesis, the Q_{n-1}^0 is f_a-adjacently f_e edges bipancyclic for $f_a + f_e = n - 3$. Let x, y be a pair of adjacently faulty vertices in F_a. Thus, $Q_{n-1}^0 - F_e - (F_a - \{x, y\})$ is bipancyclic. There exists a Hamiltonian cycle in $Q_{n-1}^0 - F_e - (F_a - \{x, y\})$. We can denote this cycle by $\langle x, w_1 \xrightarrow{P_1} b_1, y, b_2 \xrightarrow{P_2} w_2, x \rangle$. Let $|P_1|$ and $|P_2|$ denote the length of P_1 and P_2, respectively. Without loss of generality, we can assume that $|P_1| \leq |P_2|$. Thus, we will consider the following two conditions.

Case 1.1: $|P_1| \geq 1$.

Let b_1' be a black vertex of P_1 and $P(w_1, b_1')$ be the subpath between w_1 and b_1' of P_1. Applying Lemma 1, we can know that Q_{n-1}^1 is Hamiltonian laceable. There exists a Hamiltonian path $P(\phi(b_1'), \phi(w_1))$ in Q_{n-1}^1. Thus, we can construct the cycles with every even length from $2^{n-1} + 2$ to $2^{n-1} + |P_1| + 1$ with $\langle w_1 \xrightarrow{P(w_1,b_1')} b_1', \phi(b_1') \xrightarrow{P(\phi(b_1'),\phi(w_1))} \phi(w_1), w_1 \rangle$, as illustrated in Figure 1.

Let b_2' be a black vertex of P_2 and $P(w_2, b_2')$ be the subpath between w_2 and b_2' of P_2. Applying Lemma 2, we can obtain that there are two spanning disjoint paths $P(\phi(b_1), \phi(w_2))$ and $P(\phi(b_2'), \phi(w_1))$ in Q_{n-1}^1. Thus, we can construct the cycles with every even length from $2^{n-1} + |P_1| + 3$ to $2^n - 2f_a$ with $\langle w_1 \xrightarrow{P_1} b_1, \phi(b_1) \xrightarrow{P(\phi(b_1),\phi(w_2))} \phi(w_2), w_2 \xrightarrow{P(w_2,b_2')} b_2', \phi(b_2') \xrightarrow{P(\phi(b_2'),\phi(w_1))} \phi(w_1), w_1 \rangle$, as illustrated in Figure 2.

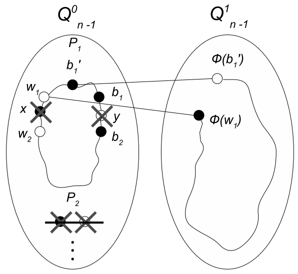

Figure 1: **Case 1.1** cycles with even length from $2^{n-1} + 2$ to $2^{n-1} + |P_1| + 1$

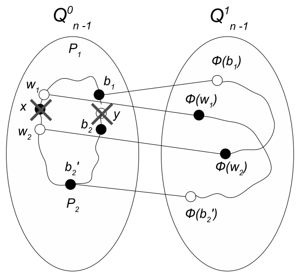

Figure 2: **Case 1.1** cycles with even length from $2^{n-1} + |P_1| + 3$ to $2^n - 2f_a$

Case 1.2: $|P_1| = 0$.

Let b_2' be a black vertex of P_2 and $P(w_2, b_2')$ be the subpath between w_2 and b_2' of P_2. Applying Lemma 1, we can know that Q_{n-1}^1 is Hamiltonian laceable. There exists a Hamiltonian path $P(\phi(b_2'), \phi(w_2))$ in Q_{n-1}^1. Thus, we can construct the cycles with every even length from $2^{n-1} + 2$ to $2^n - 2f_a$ with $\langle w_2 \xrightarrow{P_2(w_2,b_2')} b_2', \phi(b_2') \xrightarrow{P(\phi(b_2'),\phi(w_2))} \phi(w_2), w_2 \rangle$, as illustrated in Figure 3.

Case 2: $f_a^1 = 0$ and $f_a + f_e^0 \leq n - 3$.

With induction hypothesis, Q_{n-1}^0 is f_e^n edges

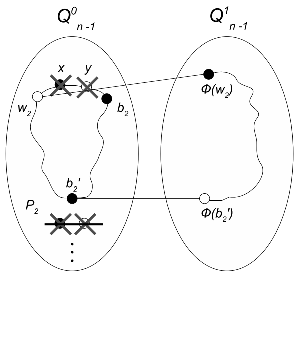

Figure 3: **Case 1.2** cycles with even length from $2^{n-1}+2$ to 2^n-2f_a

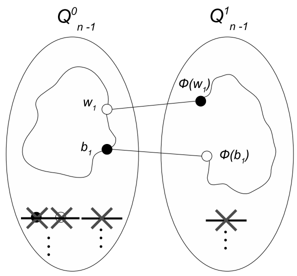

Figure 4: **Case 2** cycles with even length from $2^{n-1}-2f_a+2$ to 2^n-2f_a

f_a-adjacently bipancyclic for $f_e^0+f_a=n-3$. Thus, there exist cycles in Q^0_{n-1} with every even length from 4 to $2^{n-1}-2f_a$. Let $\langle w_1 \overset{P(w_1,b_1)}{\longrightarrow} b_1, w_1\rangle$ be the cycle with length $2^{n-1}-2f_a$ in $Q^0_{n-1}-F_a-F^0_e$. Applying Lemma 3, Q^1_{n-1} is edges bipancyclic. There exist paths $P(\phi(b_1),\phi(w_1))$ with every odd length k for $1\le k\le 2^{n-1}-1$. Therefore, we can construct the cycles with every even length from $2^{n-1}-2f_a+2$ to 2^n-2f_a with $\langle w_1 \overset{P(w_1,b_1)}{\longrightarrow} b_1, \phi(b_1) \overset{P(\phi(b_1),\phi(w_1))}{\longrightarrow} \phi(w_1), w_1\rangle$, as illustrated in Figure 4.

Case 3: $f_a^1 \ge 1$.
Since $f_a^0 \ge f_a^1 \ge 1$, $Q^1_{n-1}-F^1_a-F^1_e$ is bipancyclic. There exist cycles with every even length from 4 to $2^{n-1}-2f^1_a$ in $Q^1_{n-1}-F^1_a-F^1_e$. We will construct the cycles with every even length from $2^{n-1}-2f^1_a+2$ to 2^n-2f_a in the following cases.

Case 3.1: $n=4$.
Since $n=4$, $f_a^0=f_a^1=1$ and $f_e=0$. Applying Lemma 1, Q^0_{n-1} and Q^1_{n-1} is Hamiltonian. Let C_1 be the Hamiltonian cycle in $Q^0_{n-1}-F^0_a$. Since $f^1_a=1$, there exist two edges (b_1,w_1) and (b_2,w_2) of C_1 such that $\phi(b_1),\phi(w_1),\phi(b_2),\phi(w_2)\notin F_a$. We can denote C_1 as $\langle b_1, w_1 \overset{P(w_1,b_2)}{\longrightarrow} b_2, w_2 \overset{P(w_2,b_1)}{\longrightarrow} b_1\rangle$. Thus, $\langle b_1,\phi(b_1),\phi(w_1), w_1 \overset{P(w_1,b_2)}{\longrightarrow} b_2, w_2 \overset{P(w_2,b_1)}{\longrightarrow} b_1\rangle$ and $\langle b_1,\phi(b_1),\phi(w_1), w_1 \overset{P(w_1,b_2)}{\longrightarrow} b_2,\phi(b_2),\phi(w_2), w_2 \overset{P(w_2,b_1)}{\longrightarrow} b_1\rangle$ form the cycles with length 8 and 10, respectively. Figure 5 is illustrated for these cycles. Applying Lemma 1, we can obtain there exists a Hamiltonian cycle in Q_4 with $f_a=2$. Thus, this theorem holds for $n=4$.

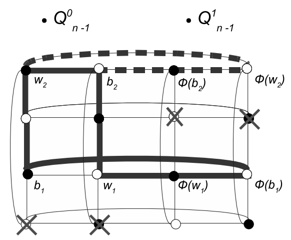

Figure 5: **Case 3.1** cycles with length 8 and 10

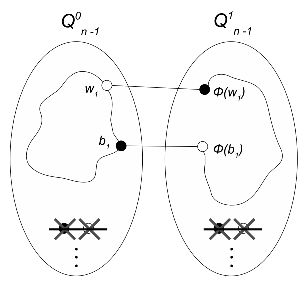

Figure 6: **Case 3.2** cycles with even length between $2^{n-1}-2f^1_a+2$ and 2^n-2f_a

Case 3.2: $n \geq 5$.

Since $f_a^0 \geq f_a^1 \geq 1$ and $n \geq 5$, $f_a^1 \leq n-4$ and $f_a^1 + f_e^1 \leq n-3$. Thus, $Q_{n-1}^1 - F_a^1 - F_e^1$ is Hamiltonian laceable and $Q_{n-1}^0 - F_a^0 - F_e^0$ is Hamiltonian by Lemma 1. Let C be the Hamiltonian cycle of $Q_{n-1}^0 - F_a^0 - F_e^0$. Let $P(w_1, b_1)$ be a path on cycle C with odd length k for $\phi(w_1), \phi(b_1) \notin F_a^1$ for $1 \leq k \leq 2^{n-1} - 2f_a^0$. Since $Q_{n-1}^1 - F_a^1 - F_e^1$ is Hamiltonian laceable, there exists a Hamiltonian path $P(\phi(b_1), \phi(w_1))$ in $Q_{n-1}^1 - F_a^1 - F_e^1$. Therefore, we can construct the cycles with every even length between $2^{n-1} - 2f_a^1 + 2$ and $2^n - 2f_a$ with $\langle w_1 \xrightarrow{P(w_1,b_1)} b_1, \phi(b_1), \xrightarrow{P(\phi(b_1),\phi(w_1))} \phi(w_1), w_1 \rangle$, as illustrated as Figure 6. $\square$

3 Conclusion

In this paper, we show that Q_n is f_a-adjacency $(n-2-f_a)$ edges bipancyclic for $0 \leq f_a \leq n-2$ and $n \geq 2$. We will investigate the adjacent vertices fault tolerance for edge-bipancyclicity of hypercube in the future.

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